

Coulomb's Law, Gauss's Law, Electric Potential, and Capacitance

Diego Juarez

Department of Physics
The University of Texas at El Paso

April 15, 2026

Outline

- 1 Coulomb's Law
- 2 Gauss's Law
- 3 Electric Potential
- 4 Capacitance
- 5 Summary

Electric Charge and Interaction

Electric charge is a fundamental property of matter. There are two kinds of charge:

- Positive charge
- Negative charge

Charges interact through the electric force:

- Like charges repel.
- Unlike charges attract.

SI unit

The SI unit of charge is the coulomb (C).

Coulomb's Law

For two point charges q_1 and q_2 separated by a distance r , the magnitude of the electric force is

$$F = k_e \frac{|q_1 q_2|}{r^2}, \quad (1)$$

where

$$k_e = \frac{1}{4\pi\epsilon_0}. \quad (2)$$

In vector form,

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12}. \quad (3)$$

Key idea

The force decreases with the square of the distance.

Superposition Principle

If many charges are present, the total electric force on a charge is the vector sum of the individual forces:

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i. \quad (4)$$

Similarly, the electric field also obeys superposition:

$$\vec{E}_{\text{net}} = \sum_i \vec{E}_i. \quad (5)$$

Electric field of a point charge

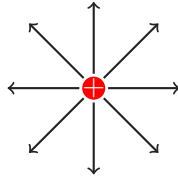
A point charge q produces an electric field

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}. \quad (6)$$

Field Lines Interpretation

Electric field lines help visualize the field:

- They point away from positive charges.
- They point toward negative charges.
- Their density indicates field strength.
- They never cross.



Electric Flux

Electric flux measures how much electric field passes through a surface:

$$\Phi_E = \int \vec{E} \cdot d\vec{A}. \quad (7)$$

For a uniform field crossing a flat surface,

$$\Phi_E = EA \cos \theta, \quad (8)$$

where θ is the angle between $\vec{E}$ and the outward normal to the surface.

Meaning

Flux is large when the field is strong and aligned with the surface normal.

Gauss's law relates the net electric flux through a closed surface to the charge enclosed:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}. \quad (9)$$

- Only the enclosed charge matters.
- Charges outside the surface do not change the net flux.
- Gauss's law is always true, but it is especially useful when symmetry is present.

Using Gauss's Law: Spherical Symmetry

For a point charge q at the center of a spherical Gaussian surface of radius r :

$$\oint \vec{E} \cdot d\vec{A} = E \oint dA = E(4\pi r^2). \quad (10)$$

Applying Gauss's law,

$$E(4\pi r^2) = \frac{q}{\epsilon_0}. \quad (11)$$

Therefore,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}. \quad (12)$$

Result

Gauss's law reproduces the electric field of a point charge.

Useful Symmetry Cases

Infinite line of charge

With linear density λ ,

$$E = \frac{\lambda}{2\pi\epsilon_0 r}. \quad (13)$$

Infinite plane of charge

With surface density σ ,

$$E = \frac{\sigma}{2\epsilon_0}. \quad (14)$$

Conductors in electrostatic equilibrium

- Electric field inside a conductor is zero.
- Excess charge resides on the surface.
- The electric field is perpendicular to the surface.

Electric Potential Difference

Electric potential difference is defined from the work done by the electric force:

$$\Delta V = - \int_a^b \vec{E} \cdot d\vec{l}. \quad (15)$$

The electric potential energy of a charge q is

$$U = qV. \quad (16)$$

Unit

Potential is measured in volts:

$$1 \text{ V} = 1 \text{ J/C}. \quad (17)$$

Potential of a Point Charge

Taking the reference $V(\infty) = 0$, the electric potential due to a point charge q is

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}. \quad (18)$$

For several point charges, superposition applies:

$$V_{\text{net}} = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i}. \quad (19)$$

Important distinction

Electric field is a vector, but electric potential is a scalar.

Relation Between Field and Potential

The electric field points in the direction of decreasing potential:

$$\vec{E} = -\nabla V. \quad (20)$$

In one dimension,

$$E_x = -\frac{dV}{dx}. \quad (21)$$

For a uniform electric field,

$$\Delta V = -Ed. \quad (22)$$

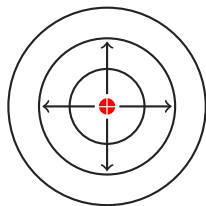
Physical meaning

A strong electric field corresponds to a rapid change of potential with position.

Equipotential Surfaces

Equipotential surfaces are surfaces where the potential is constant.

- No work is done moving a charge along an equipotential surface.
- Electric field lines are always perpendicular to equipotential surfaces.
- Conductors in electrostatic equilibrium are equipotential volumes.



Capacitance Definition

A capacitor stores electric charge and energy. Its capacitance is defined by

$$C = \frac{Q}{\Delta V}. \quad (23)$$

- Q is the magnitude of the charge on either conductor.
- ΔV is the potential difference between conductors.
- The SI unit is the farad (F).

Interpretation

Large capacitance means a device can store more charge for the same voltage.

For two parallel plates of area A separated by distance d ,

$$C = \epsilon_0 \frac{A}{d}. \quad (24)$$

This follows from

$$E = \frac{\sigma}{\epsilon_0}, \quad \Delta V = Ed, \quad Q = \sigma A. \quad (25)$$

Combining these results gives

$$C = \frac{Q}{\Delta V} = \epsilon_0 \frac{A}{d}. \quad (26)$$

A dielectric is an insulating material placed between capacitor plates.

- It reduces the effective electric field.
- It increases the capacitance.

If the dielectric constant is κ ,

$$C = \kappa \epsilon_0 \frac{A}{d}. \quad (27)$$

Key effect

Dielectrics allow more charge to be stored at the same potential difference.

Energy Stored in a Capacitor

The energy stored in a capacitor can be written in three equivalent ways:

$$U = \frac{1}{2}C(\Delta V)^2 = \frac{Q^2}{2C} = \frac{1}{2}Q\Delta V. \quad (28)$$

For a parallel-plate capacitor, the energy density in the electric field is

$$u_E = \frac{1}{2}\epsilon_0 E^2. \quad (29)$$

Idea

Capacitors store energy in the electric field between their conductors.

Summary of Main Results

Topic	Main equation
Coulomb's law	$F = \frac{1}{4\pi\epsilon_0} \frac{ q_1 q_2 }{r^2}$
Electric field	$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$
Gauss's law	$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$
Potential difference	$\Delta V = - \int \vec{E} \cdot d\vec{l}$
Point-charge potential	$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$
Capacitance	$C = \frac{Q}{\Delta V}$
Parallel-plate capacitor	$C = \epsilon_0 \frac{A}{d}$
Stored energy	$U = \frac{1}{2} C (\Delta V)^2$

Problem 1: Field of a Uniformly Charged Sphere

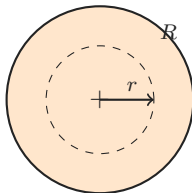
Statement. Use Gauss's law to prove the electric field of a solid sphere of radius R with uniform charge density ρ .

By spherical symmetry, $\mathbf{E}$ must be radial and have constant magnitude on a Gaussian sphere of radius r .

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(r) \oint dA$$

Since the area of a sphere is $4\pi r^2$,

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(r) 4\pi r^2.$$



Problem 1: Field of a Uniformly Charged Sphere (cont.)

Gauss's law gives

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

Therefore,

$$E(r) 4\pi r^2 = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

Now we consider two regions:

$$(1) \ r < R \qquad (2) \ r \geq R$$

Idea

For $r < R$, the Gaussian surface encloses only part of the charge.

For $r \geq R$, it encloses all the charge.

Problem 1: Field of a Uniformly Charged Sphere (inside case)

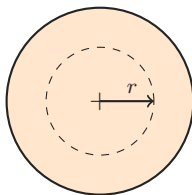
Case 1: $r < R$

The enclosed charge is the charge inside a sphere of radius r :

$$Q_{\text{enc}} = \rho \left(\frac{4}{3} \pi r^3 \right).$$

Substitute into Gauss's law:

$$E(r) 4\pi r^2 = \frac{1}{\epsilon_0} \rho \left(\frac{4}{3} \pi r^3 \right).$$



Gaussian surface inside

Problem 1: Field of a Uniformly Charged Sphere (inside case cont.)

Starting from

$$E(r) 4\pi r^2 = \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0},$$

divide both sides by $4\pi r^2$:

$$E(r) = \frac{1}{4\pi r^2} \cdot \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0}.$$

Simplifying,

$$E(r) = \frac{\rho r}{3\epsilon_0}.$$

Thus,

$$\boxed{E(r) = \frac{\rho r}{3\epsilon_0}}, \quad r < R.$$

Result

Problem 1: Field of a Uniformly Charged Sphere (outside case)

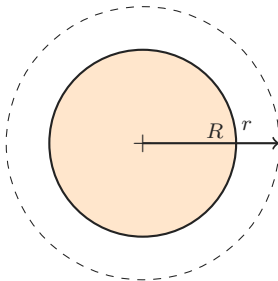
Case 2: $r \geq R$

Now the Gaussian surface encloses the entire sphere, so the enclosed charge is the total charge:

$$Q = \rho \left(\frac{4}{3} \pi R^3 \right).$$

Gauss's law becomes

$$E(r) 4\pi r^2 = \frac{Q}{\epsilon_0}.$$



Problem 1: Field of a Uniformly Charged Sphere (outside case cont.)

From

$$E(r) 4\pi r^2 = \frac{Q}{\epsilon_0},$$

divide both sides by $4\pi r^2$:

$$E(r) = \frac{Q}{4\pi\epsilon_0 r^2}.$$

Therefore,

$$\boxed{E(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}}, \quad r \geq R.$$

Using

$$Q = \rho \left(\frac{4}{3} \pi R^3 \right),$$

this is exactly the same field as that of a point charge Q located at the center.

Problem 1: Final Conclusion

The electric field of a uniformly charged solid sphere is

$$E(r) = \frac{\rho r}{3\epsilon_0}, \quad r < R$$

and

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}, \quad r \geq R$$

with

$$Q = \rho \left(\frac{4}{3} \pi R^3 \right).$$

Physical meaning

Inside the sphere, the field increases linearly with distance.

Outside the sphere, the entire charge distribution behaves exactly like a point charge at the center.

Problem 2: Capacitance of a Spherical Capacitor

Statement. Prove the capacitance of two concentric conducting spheres of radii a and b ($a < b$).

Let the inner sphere carry charge $+Q$ and the outer sphere carry charge $-Q$.

By Gauss's law, in the region $a < r < b$,

$$E(r) 4\pi r^2 = \frac{Q}{\epsilon_0},$$

so

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}.$$

The potential difference is

$$V = V(a) - V(b) = \int_a^b E(r) dr$$

and therefore

$$V = \int_a^b \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} dr.$$

Problem 2: Capacitance of a Spherical Capacitor (cont.)

Then

$$V = \frac{Q}{4\pi\epsilon_0} \int_a^b r^{-2} dr = \frac{Q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_a^b$$

so

$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right).$$

Using

$$C = \frac{Q}{V},$$

we get

$$C = \frac{Q}{\frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)} = \frac{4\pi\epsilon_0}{\frac{1}{a} - \frac{1}{b}} = 4\pi\epsilon_0 \frac{ab}{b-a}.$$

$$\boxed{C = 4\pi\epsilon_0 \frac{ab}{b-a}}$$

These four topics are closely connected:

- Coulomb's law describes the force between charges.
- Gauss's law provides a powerful way to find electric fields from symmetry.
- Electric potential gives a scalar description of electrostatics and energy.
- Capacitance explains how charge and energy are stored in conductors.

Questions?